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Team name on Kaggle leaderboard: Blood Wolf the Pojun

For each of the sections below, your reported test accuracy should approximately match the accuracy reported on Kaggle.

Perceptron

Regarding Perceptron, in the Fashion-MNIST DATASET, first, we kept learning_rate unchanged (learning_rate=0.05), and then adjusted the parameters of n_epochs (10,20,50,75). Then we find that the accuracy is the highest when n_epochs=50. The result is shown in the figure.

learning_rate=0.05			
n_epochs	Training accuracy	Validation accuracy	Test accuracy
10	83.630000	80.100000	79.740000
20	85.424000	80.680000	80.700000
50	85.264000	82.480000	81.880000
75	85.264000	82.480000	80.300000

Then, while keeping n_epochs unchanged (n_epochs=50), we adjusted the value of learning_rate (0.05, 0.005, 0.0005, 0.00005). Finally, we found that learning_rate=0.0005, the best effect. The result is shown in the figure.

n_epochs=50			
learning_rate	Training accuracy	Validation accuracy	Test accuracy
0.05	85.264000	80.300000	80.370000
0.005	85.850000	80.750000	80.620000
0.0005	87.658000	82.480000	81.880000
0.00005	87.398000	81.520000	81.260000

We also adjusted decay_rate. After repeated attempts, we found that for 2-class, when decay_rate is less than 0.95, accuracy gradually increases with the increase of decay_rate. However, when decay_rate is greater than 0.95, accuracy decreases gradually with the increase of decay_rate. For

mult-class, when decay_rate is less than 0.80, accuracy gradually increases with the increase of decay_rate. However, when decay_rate is greater than 0.80, accuracy decreases gradually with the increase of decay_rate.

Similarly, we adjusted the parameters in the same way for the RICE DATASET. Parameters of the RICE DATASET and Fashion-MNIST DATASET were finally adjusted, as shown in the figure below.

RICE DATASET

Optimal hyperparameters:	learning_rate = 0.05 n_epochs = 50 decay_rate = 0.95 (2-class); 0.80 (mult-class)
Training accuracy:	99.871689
Validation accuracy:	99.807534
Test accuracy:	99.780038

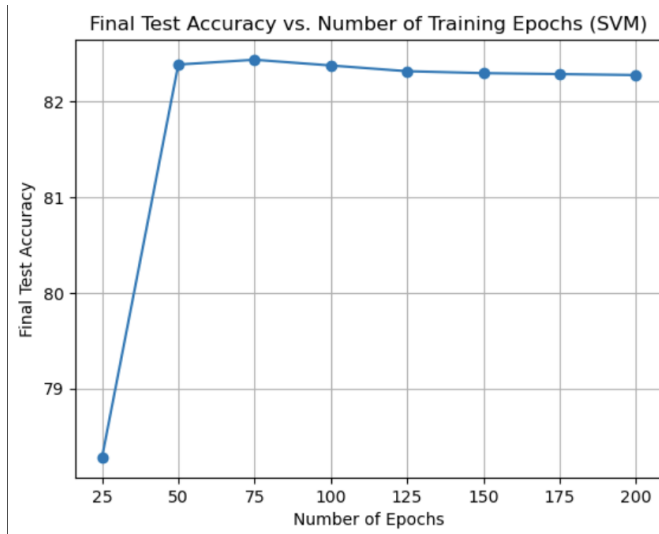
Fashion-MNIST DATASET

Optimal hyperparameters:	learning_rate = 0.0005 n_epochs = 50 decay_rate = 0.95 (2-class); 0.80 (mult-class)
Training accuracy:	87.658000
Validation accuracy:	82.480000
Test accuracy:	81.880000

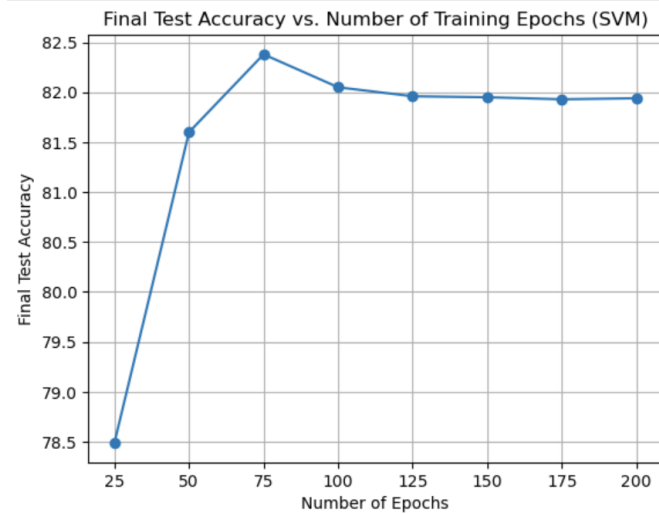
SVM

As for SVM, We first adjusted reg_const and found that it worked best when reg_const = 0.5. Then, we use python to draw graphs to predict accuracy. Four groups of pictures were drawn at learning_rate= 0.01, 0.05, 0.5 and 5. Where, the horizontal coordinate of the picture is the number of training rounds, and the ordinate is the corresponding accuracy. As shown in the four graphs below.

Learning_rate = 0.01:



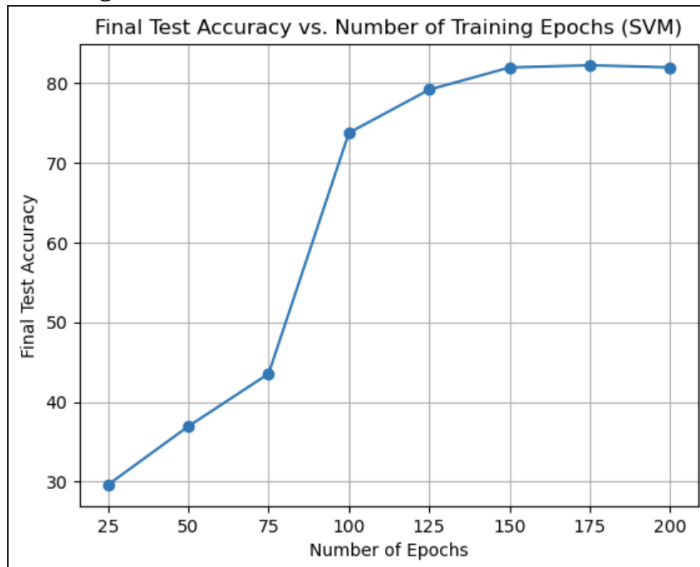
Learning_rate = 0.05



Learning_rate = 0.5:



Learning_rate = 5:



After comparing several groups of pictures, we find that when learning_rate = 0.01, the final effect is better around the 75th round. In addition, we find that the final effect is best when decay_rate = 0.95. RICE DATASET. Parameters of the RICE DATASET and Fashion-MNIST DATASET were finally adjusted, Parameters of the rice dataset and fashion - Mnist dataset were finally adjusted, as shown in the figure below.

RICE DATASET

Optimal hyperparameters:	learning_rate = 0.5 n_epochs = 500 re_const = 1 decay_rate = 0.95
Training accuracy:	99.523417
Validation accuracy:	99.450096
Test accuracy:	99.560077

Fashion-MNIST DATASET

Optimal hyperparameters:	learning_rate = 0.01 n_epochs = 75 re_const 0.5 decay_rate = 0.95
Training accuracy:	84.590000
Validation accuracy:	82.980000

Test accuracy:	82.440000
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Softmax

Once again, describe the hyperparameter tuning you tried for learning rate, number of epochs, and regularization constant. Report the optimal hyperparameter setting you found in the table below. Also report your training, validation, and testing accuracy with your optimal hyperparameter setting.

Regarding the Softmax, in the Fashion-MNIST DATASET, we first kept learning_rate unchanged (learning_rate=0.5) and re_const unchanged (re_const=0.005). Then the n_epochs parameters were adjusted (25,50,75), and it was found that the accuracy was the highest when n_epochs=50. The result is shown in the figure.

lr=0.5	re_const=0.005		
n_epochs	Training accuracy	Validation accuracy	Test accuracy
25	86.376000	85.190000	84.090000
50	86.668000	85.450000	84.310000
75	86.668000	85.420000	84.370000

Then we adjust the value of learning_rate (0.05,0.005, 0.0005) while keeping n_epochs unchanged (n_epochs=50) and re_const unchanged (re_const=0.005). Finally, we found that learning_rate=0.5, the best effect. The result is shown in the figure.

n_epochs=50	re_const=0.005		
lr	Training accuracy	Validation accuracy	Test accuracy
5	82.182000	81.200000	80.170000
0.5	86.668000	85.450000	84.310000
0.05	84.636000	83.840000	82.920000

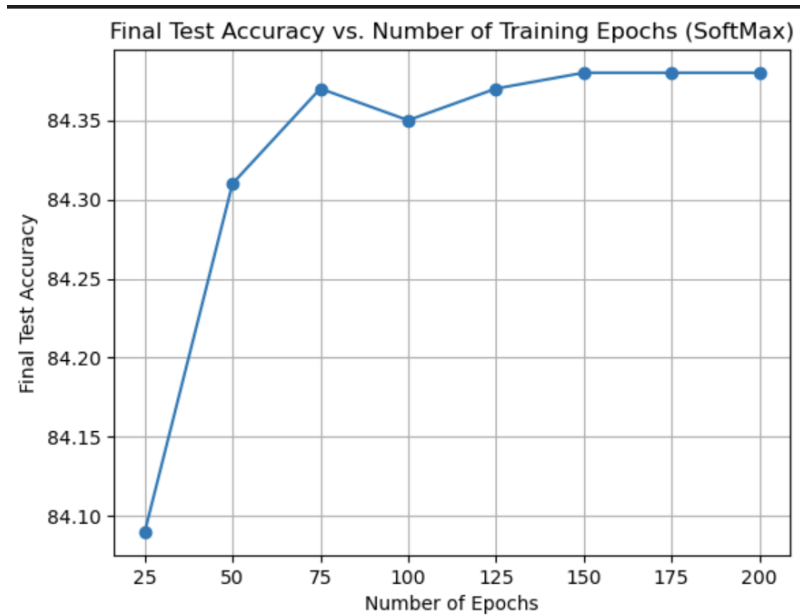
After that, while keeping n_epochs unchanged (n_epochs=50) and learning_rate unchanged (learning_rate=0.5), we adjusted re_const numerically (0.5, 0.05, 0.005) and finally found that, This works best when re_const=0.05. The result is shown in the figure.

n_epochs=50	lr=0.5		
re_const	Training accuracy	Validation accuracy	Test accuracy
0.05	84.924000	84.010000	83.150000

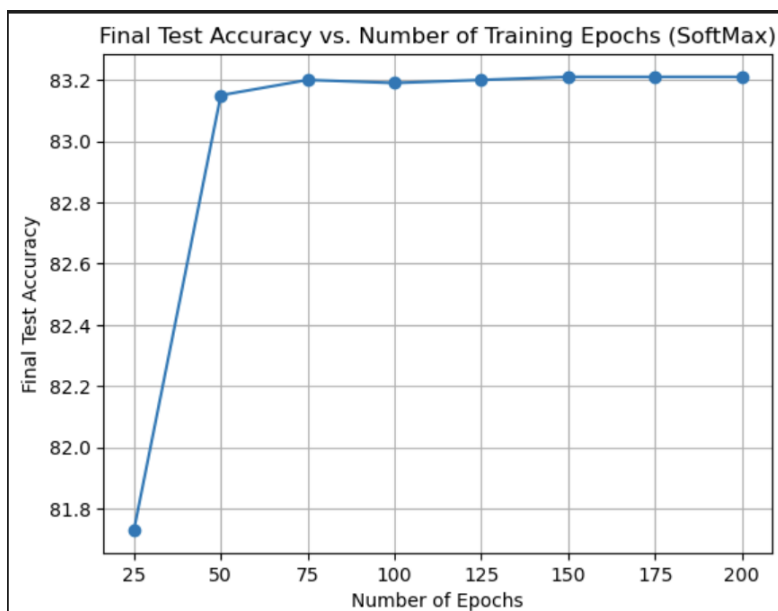
0.005	86.668000	85.450000	84.310000
0.0005	86.894000	85.390000	84.180000

In addition, we also plot some figures with different re_const to help us to find the most suitable value of re_const as follows.

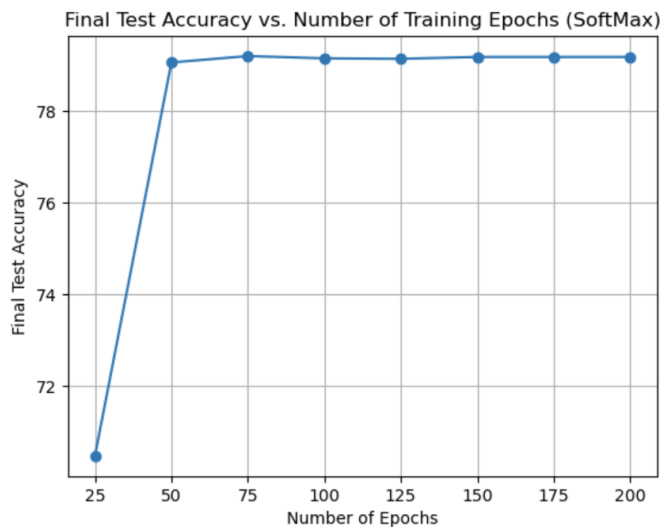
re_const = 0.005



re_const = 0.05



re_const = 0.5



We also adjusted decay_rate. After repeated attempts, we found that when decay_rate is less than 0.95, the accuracy gradually increases with the increase of decay_rate. However, when decay_rate is greater than 0.95, accuracy decreases gradually with the increase of decay_rate.

Similarly, we adjusted the parameters in the same way for the RICE DATASET. Parameters of the RICE DATASET and Fashion-MNIST DATASET were finally adjusted, as shown in the figure below.

RICE DATASET

Optimal hyperparameters:	learning_rate = 0.5 n_epochs = 100 re_const 0.05 decay_rate = 0.95
Training accuracy:	99.880854
Validation accuracy:	99.752543
Test accuracy:	99.835029

Fashion-MNIST DATASET

Optimal hyperparameters:	learning_rate = 0.5 n_epochs = 50 re_const 0.05 decay_rate = 0.95
Training accuracy:	86.668000
Validation accuracy:	85.450000

Test accuracy:	84.310000
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Logistic

Once again, describe the hyperparameter tuning you tried for learning rate, number of epochs, and threshold. Report the optimal hyperparameter setting you found in the table below. Also report your training, validation, and testing accuracy with your optimal hyperparameter setting.

Regarding hyperparameter tuning, considering the intrinsic properties of Logistic Regression, the threshold at 0.5 serves as the symmetric center of the function, making it the decision boundary between the two classes. Therefore, this parameter does not need to be adjusted and should remain at threshold = 0.5.

For tuning learning rate and number of epochs, our group initially set n_epochs to an small value (n_epochs = 50) while varying the learning rate by factors of $10^{(-1)}$ to determine its optimal range. The specific results are shown in the figure.

n_epochs=50	threshold=0.5		
learning_rate	Training accuracy	Validation accuracy	Test accuracy
5	63.972138	62.304097	63.926313
0.5	64.017964	62.331592	64.008798
0.05	68.655485	69.755293	68.930437
0.005	63.852992	62.139126	63.81332

Subsequently, we found that when learning_rate = 0.05, the performance was relatively good. Keeping learning_rate = 0.05 fixed, we then increased n_epochs and observed that when n_epochs was below 15000, increasing n_epochs led to improved performance. However, beyond n_epochs = 15000, the accuracy hardly improved any further, as illustrated in the figure.

learning_rate=0.05	threshold=0.5		
n_epochs	Training accuracy	Validation accuracy	Test accuracy
50	68.655485	69.755293	68.930437
5000	71.450829	70.195216	70.992576
10000	79.946843	80.863349	79.763541

15000	99.917514	99.862524	99.862524
20000	99.926679	99.890019	99.862524

Finally, we explored the effect of decay_rate. We found that as decay_rate decreased, the accuracy also dropped significantly, as shown in the figure.

learning_rate=0.05	threshold=0.5	n_epochs=15000	
decay_rate	Training accuracy	Validation accuracy	Test accuracy
1	99.917514	99.862524	99.862524
0.99	99.395106	99.285125	99.31262
0.98	98.102832	98.377784	98.047842
0.97	78.929521	78.526258	79.186142

In conclusion, the optimal hyperparameters were determined as learning_rate = 0.05, threshold = 0.5, n_epochs = 15000, and no decay_rate. The final results are summarized in the table below.

RICE DATASET

Optimal hyperparameters:	learning_rate = 0.05 n_epochs = 15000 threshold = 0.5
Training accuracy:	99.917514
Validation accuracy:	99.862524
Test accuracy:	99.862524